Lesson 1.1 - Functions and Transformations

Course: Advanced Functions

Learning Goals

- Use function notation and evaluate functions.
- Identify domain and range.
- Describe transformations from a parent function.

Key Notes

A function assigns exactly one output to each permitted input. Function notation such as $f(x)$ names the rule and the input. For $y = a f(k(x-d)) + c$: d produces a horizontal translation, c a vertical translation, a affects vertical stretch/compression and reflection, and k affects horizontal stretch/compression and reflection.

Worked Example 1

Let $f(x) = 2x^2 - 3$. Then $f(4) = 2(4)^2 - 3 = 29$. If $f(x) = 15$, solve $2x^2 - 3 = 15$, so $x^2 = 9$ and $x = \pm 3$.

Worked Example 2

Compared with $y = x^2$, the graph $y = -2(x-3)^2 + 5$ is reflected in the x-axis, vertically stretched by a factor of 2, translated 3 units right, and 5 units up.

Teacher Reminder

Horizontal transformations often feel backwards because they occur inside the function. Check the transformed x-coordinate rather than guessing.

Practice

- For $f(x)=3x-7$, calculate $f(-2)$.
- State the transformations from $y=x^2$ to $y=0.5(x+4)^2-1$.
- Determine the domain and range of $y=\sqrt{x-2}+3$.
- Sketch $y=-|x+1|+4$ from the parent $y=|x|$.